

Cyber Security

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CHAPTER-4

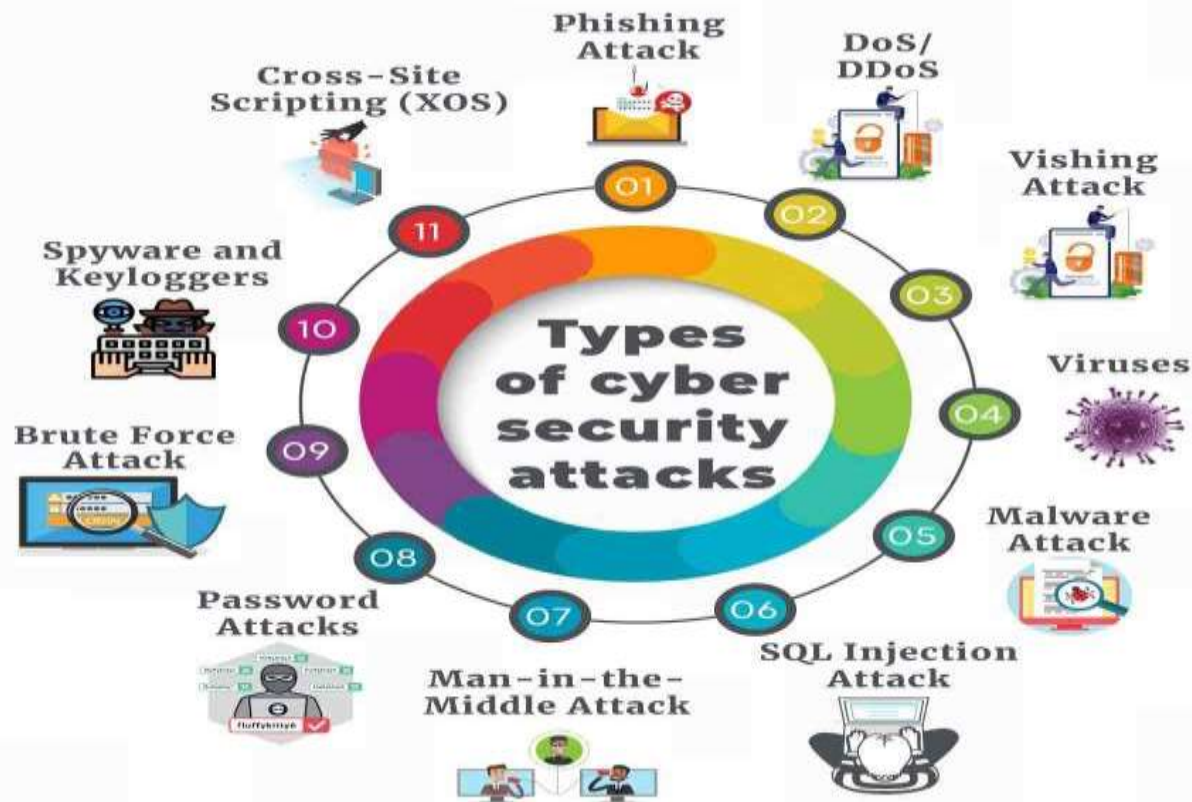
Introduction to Cyber Crime and law



Cyber Crimes

- Cybercrimes are defined as illegal activities in which a computer or network serves as the crime's source, instrument, destination, or site.
- To put it simply, cybercrimes are any illegal conduct including a network of information technology, illicit access, unlawful interception, tampering with data, and system interference, device abuse, counterfeiting, and electronic fraud.
- Cybercrime, the use of a computer as an instrument to further illegal ends, such as committing fraud, trafficking in child pornography and intellectual property, stealing identities, or violating privacy.

Types of Cyber crimes



Types of Cyber crimes

Two Main Types of Cyber Crimes

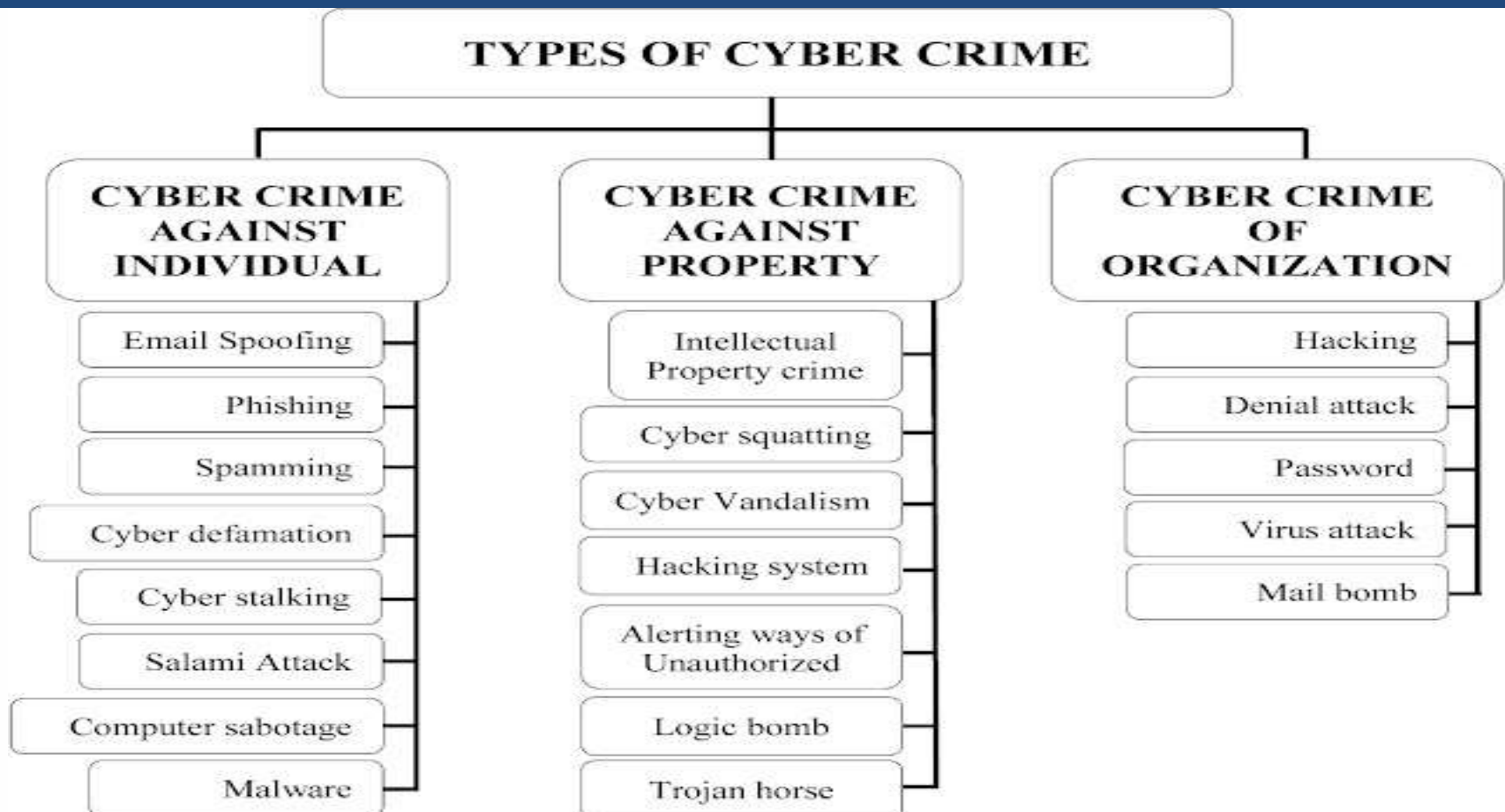
Targeting computers

This type of cybercrimes includes every possible way that can lead to harm to computer devices for example malware or denial of service attacks.

Using computers

This type includes the usage of computers to do all the classifications of computer crimes.

Types of Cyber crimes



Types of Cyber crimes

- **Cybercrime against individual**

The term cybercrime against the individual refers to those criminal offences which are committed against an individual. Such cybercrime affects the individual's personality.





Types of Cyber crimes

- **Electronic mail (E-Mail) Spoofing:** E-mail spoofing is the forgery of an e-mail header so that the message appears to have originated from someone or somewhere other than the actual source.
- **Phishing:** It is an act of criminally fraudulent process of attempting to acquire sensitive information such as usernames, passwords and credit card details by masquerading as a trustworthy entity in an electronic communication.
- **Spear Phishing:** Spear phishing is an email or electronic communications scam targeted towards a specific individual, organization or business.
- **Vishing:** Vishing is the criminal practice of using social engineering over the telephone system, most often using features facilitated by VoIP, to gain access to personal and financial information from the public for the purpose of financial reward. The term is a combination of V-voice and Phishing.
- **Smishing:** The name is derived from “SMs PhISHING”. Smishing uses cell phone text messages to deliver a lure message to get the victim to reveal his/her personal information.



Types of Cyber crimes

- **Spamming:** Spam is sending undesired junk emails and commercial messages over internet. People who create electronic spam are called 'spammers'.
- **Cyber defamation:** The act of defaming, insulting, offending or otherwise causing harm through false statements about a person, company or nation etc. through internet.
- **Cyber stalking and harassment:** Cyber stalking refers to the use of internet and/or other electronic communication devices to stalk another person. It involves repeatedly harassing or threatening an individual via the internet or other electronic means of communication.
- **Computer Sabotage:** Computer sabotage involves deliberate attacks intended to disable computers or networks.
- **Pornographic offenses:** Cyber pornography is the act of using cyberspace to create, display, distribute, import, or publish pornography or obscene materials. Child pornography means any visual depiction, including but not limited to the following: Any photograph that can be considered obscene and/or unsuitable for the age of child viewer.



Types of Cyber crimes

- **Password Sniffing:** Password sniffers are the programs that can monitor and records passwords that are used or broadcasted on a computer or network interface. It listens to all incoming and outgoing network traffic and records any instance of a data packet that contains a password.
- **Identity Theft:** it is a fraud involving another person's identity for an illicit purpose. It occurs when a criminal uses someone else's identity for his/her own illegal purpose. Phishing and identity theft are related offenses.

Types of Cyber crimes

Classifications of Cybercrimes

- Cybercrime against property

The second category of cybercrime is that of cybercrimes which affects person's property. These cybercrimes are known as cybercrimes against property.

1. Credit cards frauds
2. Intellectual property (IP) crimes
3. Internet time theft



Types of Cyber crimes

- **Credit cards frauds:** Credit card (or debit card) fraud is a form of identity theft that involves an unauthorized person taking of another's credit card information for the purpose of charging purchases to the account or removing funds from it. The purpose may be to obtain goods without paying, or to obtain unauthorized funds from an account.
- **Intellectual property (IP) crimes:** Basically, IP crimes include software piracy, copyright infringement, trademarks violations, theft of computer source code, etc.
- **Internet time theft:** Occurs when an unauthorized person uses the internet hours paid for by another person. The person who gets access to someone else's ISP user ID and password, either by hacking or by gaining access to it by illegal means, uses it to access the internet without the other person's knowledge.

Types of Cyber crimes

- **Cybercrime against organization**

There are certain offences done by group of persons intending to threaten the international governments or firm, company, group of Individuals by using internet facilities. These cybercrimes are known as cybercrimes against Organization.

1. Unauthorized accessing of computer
2. Password sniffing
3. Denial-of-Service (DoS) attacks
4. Virus attack
5. E-mail bombing
6. Salami attack
7. Logic bomb
8. Trojan horse
9. Data diddling
10. Industrial spying/Industrial espionage
11. Computer network intrusions
12. Software piracy



Types of Cyber crimes

- **Unauthorized accessing of computer:** Hacking is one method of doing this. Hackers make use of the weaknesses and loop holes present in systems to destroy data and steal important information from victim's computer. Every act committed toward breaking into a computer and/or network is hacking and it is an offense.
- **Password sniffing:** Password sniffers are the programs that can monitor and records passwords that are used or broadcasted on a computer or network interface. It listens to all incoming and outgoing network traffic and records any instance of a data packet that contains a password.
- **Denial-of-Service (DoS) attacks:** A denial-of-service attack (DoS attack) is the intrusion into a system by disabling the network with the intent to deny service to authorized users. Attackers achieve this by flooding a network with more traffic than it can handle.
- **Virus attack:** Computer virus is a program that can 'infect' legitimate programs by modifying them to include a possibly 'evolved' copy of itself. Virus spread themselves, without the knowledge or permission of the users, to potentially large number of programs on many machines.



Types of Cyber crimes

- **E-mail bombing/mail bombs:** E-mail bombing refers to sending a large number of E-mails to the victims to crash victim's E-Mail account or to make victim's mail servers crash.
- **Salami attack/Salami technique:** Salami attack is when small attacks add up to one major attack that can go undetected due to the nature of this type of cybercrime. Salami attacks are used for committing financial crimes and are difficult to detect and trace. For example, a fraud activity in a bank, where an employee steals a small amount of funds from several accounts, can be considered a salami attack.
- **Logic bomb:** These are event dependent programs. This implies that these programs are created to do something only when a certain event (known as a trigger event) occurs. For example, some viruses may be termed logic bombs because they lie dormant all through the year and become active only on a particular date (like the Chernobyl virus).
- **Trojan horse:** Trojan horse is a program in which malicious or harmful code is contained inside apparently harmless programming or data in such a way that it can get control and cause harm.



Types of Cyber crimes

- **Industrial spying/industrial espionage:** It is the illegal practice of investigating competitors to gain a business advantage. The target of investigation might be a trade secret such as product specification or formula or information about business plans.
- **Computer network intrusions:** An intrusion detection system (IDS) inspects all inbound and outbound network activity and identifies suspicious patterns that may indicate a network or system attack from someone attempting to break into or compromise a system.
- **Software piracy:** Theft of software through the illegal copying of genuine programs is known as software piracy.
 - Examples:
 - ☐ End user copying
 - ☐ Hard disk loading with illicit means
 - ☐ Counterfeiting
 - ☐ Illegal downloads from the Inter

Types of Cyber crimes

4 Cybercrime against society

- Those cybercrimes which affects the society interest at large are known as cybercrimes against society.
 1. Forgery
 2. Cyber terrorism
 3. Web jacking



Types of Cyber crimes

- **Forgery:** The act of forging something, especially a document or object for the purpose of fraud or deception.
 - Examples:
 - ☐ Counterfeit currency notes
 - ☐ Postage and revenue stamps
 - ☐ Mark sheets or even degree certificates can be forged using sophisticated computers, printers and scanners.
- **Cyber terrorism:** Cyber terrorism is the convergence of cyberspace and terrorism. It is the activity carried out by terrorist on the internet to disrupt large number of system networks with the means of computer virus.
- **Web jacking:** Web jacking occurs when someone forcefully takes control of a website. The first stage of this crime involves “password sniffing”. The actual owner of the website does not have any more control over what appears on that website.



Hacking

- Every act committed toward breaking into a computer and/or network is hacking.
- A commonly used hacking definition is the act of compromising digital devices and networks through unauthorized access to an account or computer system. Hacking is not always a malicious act, but it is most commonly associated with illegal activity and data theft by cyber criminals.
- Hacking in cyber security refers to the misuse of devices like computers, smartphones, tablets, and networks to cause damage to or corrupt systems, gather information on users, steal data and documents, or disrupt data-related activity.



Types of Hacking/Hackers

Types Of Hackers



Grey Hat Hackers



Black Hat Hackers



White Hat Hackers



Types of Hacking/Hackers cont...

- **Black Hat Hackers:** Black hat hackers are the "bad guys" of the hacking scene. They go out of their way to discover vulnerabilities in computer systems and software to exploit them for financial gain or for more malicious purposes, such as to gain reputation, carry out corporate espionage, or as part of a nation-state hacking campaign. These individuals' actions can inflict serious damage on both computer users and the organizations they work for. They can steal sensitive personal information, compromise computer and financial systems, and alter or take down the functionality of websites and critical networks.
- **White Hat Hackers:** White hat hackers can be seen as the "good guys" who attempt to prevent the success of black hat hackers through proactive hacking. They use their technical skills to break into systems to assess and test the level of network security, also known as ethical hacking. This helps expose vulnerabilities in systems before black hat hackers can detect and exploit them.



Types of Hacking/Hackers cont...

- **Grey Hat Hackers:** Grey hat hackers sit somewhere between the good and the bad guys. Unlike black hat hackers, they attempt to violate standards and principles but without intending to do harm or gain financially. Their actions are typically carried out for the common good. For example, they may exploit a vulnerability to raise awareness that it exists, but unlike white hat hackers, they do so publicly. This alerts malicious actors to the existence of the vulnerability.

Hierarchy of Contemporary Cyber-Criminals

- **Hierarchy of Contemporary Cyber-Criminals**

There are four general categories of Cyber-criminals.

1. Script Kiddies
2. Cyberpunks
3. Hackers/crackers
4. Cyber-criminal organizations

- **Script Kiddies:** Also known as Skidiots, Skiddie or Victor Skill Deficiency (VSD). These are the lowest form of cyber-criminal. Inexperienced hackers who employ scripts or other programs authored by others to exploit security vulnerabilities or otherwise compromise computer systems.
 - [?] Technologically the least sophisticated of all cyber-criminals.
 - [?] Generally not capable of writing their own programs.



Hierarchy of Contemporary Cyber-Criminals

- **Cyberpunks:** Refers to individuals that create havoc via the internet. The term was initially used to refer to an emerging genre which marries science fiction, information technology, and radical change in social order.
- **Hackers/Crackers:** Sophisticated computer criminals that are capable of programming, writing code and breaching complex systems are categorized as hackers or crackers. Hackers are those individuals who identify and exploit system vulnerabilities but who lack economic motivation.

Crackers are those sophisticated users who employ their knowledge for personal gain.

- **Cyber-criminal organizations:** Groups comprised of criminally minded individuals who have used the internet to communicate, collaborate, and facilitate cyber_crime. Criminal hackers are those who target data which is valuable i.e. for example trade secrets, proprietary data, credit card data etc which may be used to further other criminal activity.



Prevention from Getting Hacked

- Software Update
- Use Unique Passwords for Different Accounts
- HTTPS Encryption
- Avoid Clicking on Ads or Strange Links
- Change the Default Username and Password on Your Router and Smart Devices
- Download from First-party Sources
- Install Antivirus Software
- Use a VPN
- Do Not Login as an Admin by Default
- Use a Password Manager
- Use Two-factor Authentication
- Brush Up on Anti-phishing Techniques



ATTACK VECTORS

- An attack vector is a path or means by which a hacker can gain access to a computer or network server in order to deliver a malicious payload or malware.
- Attack vectors enable hackers to exploit system vulnerabilities, including the human element.
- Attack vectors include viruses, e-mail attachments, Web pages, pop-up windows, instant messages, chat rooms, and deception. All of these methods involve programming except deception
- To some extent, firewalls and anti-virus software can block attack vectors. But no protection method is totally attack-proof.
- A defense method that is effective today may not remain so for long, because hackers are constantly updating attack vectors, and seeking new ones to gain unauthorized access to computers and servers.
- The most common malicious payloads are viruses, Trojan horses, worms, and spyware.

ATTACK VECTORS

Different ways to launch attack vectors

1. Attack by E-Mail
2. Attachments (and other files)
3. Attack by deception
4. Hackers
5. Heedless guests (attack by webpage)
6. Attack of the worms
7. Malicious macros
8. Foistware (sneakware)
9. Viruses





ATTACK VECTORS

Attack by E-mail: The hostile content is either embedded in the message or linked to by the message. Sometimes attacks combine the two vectors, so that if the message does not get you, the attachment will.

- Spam is almost always carrier for scams, fraud, dirty tricks, or malicious actions of some kind.
- Any link that offers something “free” or tempting is a suspect.

Attachments (and other files): Malicious attachments install malicious computer code.

- The code could be a virus, Trojan horse, Spyware, or any other kind of malware.
- Attachments attempt to install their payload as soon as you open them.

ATTACK VECTORS

Attack by deception: Deception is aimed at the user/operator as a vulnerable entry point.

- It is not just malicious computer code that one needs to monitor.
- Fraud, scams, hoaxes, and to some extent spam, not to mention viruses, worms and such require the unwitting cooperation of the computer's operator to succeed.
- Social engineering and hoaxes are other forms of deception that are often an attack vector too.

Hackers: Hackers/crackers are a formidable attack vector because, unlike ordinary malicious code, people are flexible and they can improvise.



ATTACK VECTORS

Heedless guests (attack by webpage): Counterfeit websites are used to extract personal information. Such websites look very much like the genuine websites they imitate.

- One may think he/she is doing business with someone you trust.

However he/she is really giving their personal information, like address, credit card number, and expiration date.

- They are often used in conjunction with Spam, which gets you there in the first place.
- Pop-up webpages may install Spyware, Adware or Trojans.

Attack of the worms: Many worms are delivered as E-Mail attachments, but network worms use holes in network protocols directly. Any remote access service, like file sharing, is likely to be vulnerable to this sort of worm. In most cases, firewalls will block system worms.



ATTACK VECTORS

- Many of these system worms install Trojan Horses. Next they begin scanning the internet from the computer they have just infected, and start looking for other computers to infect.
- If the worm is successful, it propagates rapidly.
- The worm owner soon has thousands of “zombie” computers to use for more mischief.

Malicious macros: Microsoft Word and Microsoft Excel are some of the examples that allow macros. A macro does something like automating a spreadsheet, for example.

- Macros can be used for malicious purposes. All internet services like instant messaging, Internet Relay Chart (IRC), and P2P file_x0002_sharing networks rely on cozy connections between the computer and other computers on the internet.
- If one is using P2P software then his/her system is more vulnerable to hostile exploits.



ATTACK VECTORS

Foistware (sneakware): Foistware is the software that adds hidden components to the system on the sly.

- Spyware is the most common form of foistware.
- Foistware is quasi-legal software bundled with some attractive software.
- Sneak software often hijacks your browser and diverts you to some “revenue opportunity” that the foistware has set up.

Viruses: These are the malicious computer codes that hitch a ride and make the payload.

- Now-a-days, virus vectors include E-Mail attachments, downloaded files, worms, etc.



cyber space and criminal behaviour

- Cyberspace refers to the virtual space that provides the infrastructure, electronic medium and related elements necessary for online global communication.
- It can be thought of as the second life space where human beings operate for social interactions, entertainment, business operations as well as for personal activities and interests.
- The term cyberspace is derived from the word cybernetics which in turn is extracted from ancient Greek word kubernētēs, that refers to steersman or to give direction.
- The term cyberspace first came into existence in various contexts in visual arts and science fiction during 1940, 1960 and 1984.
- However, the first reference was made by the founder of Electronic Frontier Foundation, in the year 1990 and later in 1991 by Mr. Benedict, which is close to the existing relationship of computer and telecommunication systems.



cyber space and criminal behaviour

- The virtual library of information offers required information on any topic at any point of time and cyberspace acts as the informational resource now-a-days.
- Entertainment and social networking play a major role in cyberspace as the cyberspace has been evolving as a great medium to connect people these days.
- The disadvantages are due to this great medium of connectivity, as it leads to spamming, theft of information and threats etc.
- A cybercriminal is an individual who commits cybercrimes, where he/she makes use of the computer either as a tool or as a target or as both.
- Cybercriminals tries to use the computers in three broad ways.
- Firstly, they use the computer as their target for attacking other people's computers for the purpose of fulfilling their malicious activities like spreading viruses, data theft, identity theft, etc.



cyber space and criminal behaviour

- Secondly, they use the computer as their weapon for the purpose of carrying out conventional crime like spam, fraud, illegal gambling, etc.
- Thirdly, they use the computer as their accessory for the purpose of saving stolen or illegal data.
- Thus cyberspace provides a platform for all criminal activities.
- The advantages of cyberspace include
 - i. Informational resources
 - ii. Entertainment
 - iii. Social networking



Clarification of terms

- **Computer Crime:** A general term that has been used to denote any criminal act which has been facilitated by computer use. Such generalization has included both Internet and non-Internet activity. Examples include theft of components, counterfeiting, digital piracy or copyright infringement, hacking, and child pornography.
- **Definition of Cybercrime:** Cybercrime is any illegal behavior, directed by means of electronic operations, that targets the security of computer systems and the data processed by them.



Traditional Problems Associated with Computer Crime

- The traditional problems associated with computer crime are as follows:
 1. Physicality and Jurisdictional Concerns
 2. Perceived Insignificance, Stereotypes, and Incompetence
 3. Prosecutorial Reluctance
 4. Lack of Reporting
 5. Lack of Resources
 6. Jurisprudential Inconsistency



Traditional Problems Associated with Computer Crime

- **Physicality and Jurisdictional Concerns**

Individuals sitting at their desk can enter various countries without the use of passports or documentation.

For successful prosecution it is necessary to get the specification of the crime scene i.e.,

1. Where did the crime actually occur?
2. Which laws apply?
3. Which agency is responsible for the investigation of a particular incident?
4. Which agency has primary jurisdiction over the thief?



Traditional Problems Associated with Computer Crime

- **Perceived Insignificance, Stereotypes, and Incompetence**
- Investigators and administrators have displayed great reluctance to pursue computer criminals.
- A lack of knowledge coupled with general apathy toward cyber_x0002_criminality has resulted in an atmosphere of indifference.
- In a study conducted by the department of justice, computer crime investigators recognized the threat posed by employees and insiders.
- Thus, timely detection of these individuals who are most trusted employees with authorized access is often unlikely.
- Two-thirds of all agencies dealt with computer-related incident.
- Most viewed “harassment/stalking” via the internet as most prevalent of calls for assistance with child pornography a close second. Other crimes reported are forgery, counterfeiting, identity theft, e-commerce fraud etc.



Traditional Problems Associated with Computer Crime

- **Perceived Insignificance, Stereotypes, and Incompetence**
- Even in situations where law enforcement authorities recognize the insidious nature of computer or cybercrime, many do not perceive themselves or others in their department to investigate such criminal activity.
- Out of 34.4% of agencies, at least one individual had taken training, only 18.8% felt that person can investigate computer related crime and 12.3% is capable of forensic examinations.
- 70% of those who indicated that they received training were characterized as “basic”, “general”, or “introductory”.



Traditional Problems Associated with Computer Crime

- **Prosecutorial Reluctance**
- Like their law enforcement counterparts, prosecutors across the country lack sufficient knowledge and experience to effectively prosecute computer crime.
- In addition, many do not perceive electronic crime as serious and often grant it the lowest priority.
- Even those jurisdictions which have granted electronic crime high priority are often thwarted in their efforts by a lack of cooperation in extradition requests, the victim's reluctance to prosecute, the labor-intensive nature of case preparation, and/or the lack of resources for offender tracking.



Traditional Problems Associated with Computer Crime

- **Lack of Reporting**
- Early studies indicate that only 17% of victimizations were reported to law enforcement authorities.
- One of the primary reasons that businesses fail to report computer intrusions is their need to assure consumers of data security.
- In addition, many corporations are uncomfortable with the release of information to any entity, including law enforcement, and want to maintain control of the investigation of all times. Thus, they choose to handle things internally.
- A further reason that companies do not report is the perception that reporting will not result in capture or identification of a suspect.
- Intrusions are detected long after the violation occurred, making investigations more difficult.



Traditional Problems Associated with Computer Crime

- **Lack of Resources**
- Traditional budget constraints.
- Nature of technology - Changes in the technology requires frequent training and updation.
- Cost of training - Extremely expensive training which is out of reach for many agencies. Cost of additional personnel for every officer transferred to technology crime, another must be recruited, hired, and trained to take his/her place.
- Cost of hardware.
- Cost of software.
- Cost of laboratory.
- Inability to compete with private industry.



Traditional Problems Associated with Computer Crime

- **Jurisprudential Inconsistency**
- The Supreme Court has remained resolutely averse to deciding matters of law in the newly emerging sphere of cyberspace.
- They have virtually denied cert on every computer privacy case to which individuals have appealed and have refused to determine appropriate levels of Fourth amendment protections of individuals and computer equipment.
- As such, the country is remarkably divided on fundamental elements of law – establishing a legality standard of behaviour in one jurisdiction which negates or supersedes the standard in another..

Incident Response

The Six Steps of Incident Response

Preparation

1



Analysis

3



Eradication

5



2

Detection



4

Containment



6

Recovery



Incident Response

1) Preparation

- Preparation is crucial to effective incident response. Even the best Cyber Security Incident Response Team (CSIRT) cannot effectively respond to an incident without predetermined instructions.
- Preparedness involves:
 - Design, development, training, and implementation of enterprise-wide IR plan
 - Creating communication guidelines to enable seamless communication during and after an incident
 - Conducting cyber simulation exercises to evaluate the effectiveness of incident response plan



Incident Response

2) Detection

- The objective of this phase is to monitor networks and systems to detect, alert, and report on potential security incidents.
- Adopt cyber threat intelligence (CTI) capabilities to develop a comprehensive cyber monitoring program and to support ongoing monitoring and detection
- Conduct cyber compromise assessments to detect unknown compromises



Incident Response

3) Analysis

- The majority portion of the efforts to properly understand the security incident take place during this step. It involves:
- Gathering information and then prioritizing individual incidents and steps for a response.
- Forensic preservation and analysis of data to determine the extent and impact of the incident.



Incident Response

During the event of an incident, the incident response team should focus on three areas:

- **Endpoint Analysis**

Determine tracks left behind by the malicious actor.

Analyze a bit-for-bit copy of systems to determine what occurred on a device during the incident.

- **Binary Analysis**

Analyze malicious tools or binaries used by the malicious actor and document the functionalities of those programs. The analysis can be performed through Behaviour Analysis or Static Analysis.

- **Enterprise Hunting**

Analyze existing systems and event logs to determine the scope of the incident.

Document all the compromised systems, devices, and accounts.



Incident Response

4) Containment

- This is the most critical stage of incident response. The strategy for containing an incident is based on the intelligence and indicators of compromise gathered during the analysis phase. The security team should focus on taking risk-mitigating actions to prevent further impact and damage to the organization.
- Coordinated Shutdown: Once identifying the compromised systems, perform a coordinated shutdown of these devices. The IR team should be instructed to ensure proper timing.
- Wipe and Rebuild: Wipe the compromised systems and rebuild the operating systems from scratch. Change the login credentials of all the compromised accounts.



Incident Response

5) Eradication

Once you have identified domains or IP addresses leveraged by the malicious actors for command and control, issue 'threat mitigation requests' to block the communication from all channels connected to these domains. The IR team should remove the known existing threats from the networks.



Incident Response

6) Recovery

- Develop a near-term remediation strategy and roadmap
- Focus on resuming normal business operations
- Develop a long-term risk mitigation strategy
- Document the incident to improve the IR plan and update security measures to avoid such incidents in future



Digital Forensics

- Digital forensics is the process of storing, analyzing, retrieving, and preserving electronic data that may be useful in an investigation.
- Digital Forensics is the preservation, identification, extraction, interpretation and documentation of computer evidence which can be used in the court of law.
- Technically, the term computer forensics refers to the investigation of computers. Digital forensics includes not only computers but also any digital device, such as digital networks.



COMPUTER LANGUAGE

- Computers are the mechanism through which raw information (i.e., data) is processed.
- Although raw data may seem complex to understand, the structure of data is actually very basic, and is based on a binary language.
- The smallest piece of data is called a bit.
- Each bit has two possible electrical states, on (1) or off (0).
- Thus, raw data is a series of 1s and 0s. Of course, raw data is difficult to interpret by users, so computers group bits together to provide identifiable meaning.
- The smallest such grouping occurs when eight bits are combined to form a byte.
- Each byte of data represents a letter, number, or character. Foreexample, the raw data sequence of 01000001 appears to the user as the capital letter “A.”
- As stored information has increased, the data capacity of computers is also increased from kilobytes (KB) to megabytes (MB) to gigabytes (GB), terabytes (TB).



Network Language

- Few most commonly used terms in network language are as follows:

1. TCP/IP
2. IMAP
3. POP
4. Routers
5. Hubs
6. Packets
7. Cookies
8. DNS



Network Language

TCP/IP

- TCP/IP stands for Transmission Control Protocol/Internet Protocol.
- It refers to the suite of protocols that define the Internet.
- TCP is a method of communication between programs which enables a bit-stream transfer of information.
- Originally proposed and designed as the standard protocol for ARPANet, but now TCP/IP software is available for every major kind of computer operating system.
- Luckily, it is now built into many of the most common operating systems.

Network Language

IMAP

- IMAP stands for Internet Message Access Protocol.
- It is an internet standard protocol used by email clients to retrieve email messages from a mail server over a TCP/IP connection.
- E-mail stored on an IMAP server can be manipulated from anywhere without the need to transfer messages or files back and forth between the computers.

Routers

- Routers are defined as special-purpose computers that handle the connection between two or more networks.
- Routers spend all their time looking at the destination addresses of the packets passing through them and deciding which route to send them on.

Network Language

POP

- POP stands for Post Office Protocol.
- Post Office Protocol is a standard mail protocol used to receive emails from a remote server to a local email client.
- It allows you to download email messages on your local computer and read them even when you are offline.
- It was designed to support offline/local email processing.
- Once the messages are downloaded, they are deleted from the mail server.
- This mode of access is not compatible with access from multiple computers.

Network Language

Hubs

- Hub is used for connecting multiple computers or segments of a LAN.
- Hubs are central switching devices for communications lines in a star topology.
- Hubs may be added to bus topologies, for example, a hub can turn an Ethernet network into a star topology to improve troubleshooting.

Packets

- Packets are the basic units of communication over a TCP/IP network.
- They are defined as units of data exchanged between host computers.
- A packet is a string of bits divided into three main sections:
 1. A set of headers
 2. The payload, the actual data being transmitted
 3. The trailer, sometimes called the footer



Network Language

Cookies

- Cookies are small pieces of information that an HTTP server sends to the individual browser upon the initial connection.
- Not all browsers support cookies. However, most popular browsers such as MS Internet Explorer 3.0 or higher and Netscape Navigator 2.0 and higher.
- Cookies might contain information such as login or registration information, online “shopping cart” information, user preferences, and so on.
- When a server receives a request from a browser that includes a cookie, the server is able to use the information stored in the cookie.
- Cookies do not steal information. They simply act as storage platforms for information that a user has supplied.



Network Language

DNS

- DNS stands for Domain Name System.
- Domain Name System is a hierarchical and decentralized naming system for computers, services, or other resources connected to the Internet or a private network.
- DNS eases the translation of IP addresses through the utilization of hierarchical principles.
- Traditional top-level domain names include com (commercial organization), edu (educational institutions), gov (government organizations), org (nonprofit organizations), and net (Internet access providers).

× ○ DIGITAL LEARNING CONTENT



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